

GENERAL CHEMISTRY II LAB MANUAL



General Chemistry II Lab Manual

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Licensing

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CHAPTER OVERVIEW

Safety

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- 0.2: Emergency Procedures
- 0.3: Safety Dos and Don'ts
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0.1: Chemicals to be used

IMPORTANT NOTICE

Reagents used in CHE 141 are listed below. Safety Data Sheets for these reagents are on file in the lab. **IF YOU HAVE A MEDICAL CONDITION INCLUDING CURRENT PREGNANCY (OR YOU BECOME PREGNANT) THAT COULD AFFECT YOUR FULL PARTICIPATION IN THIS COURSE, PLEASE IMMEDIATELY CONSULT YOUR PERSONAL PHYSICIAN WITH THIS INFORMATION TO DETERMINE HOW EXPOSURE TO THESE CHEMICALS MIGHT**

AFFECT YOUR CONDITION AND THE POTENTIAL FOR HARMFUL EXPOSURE. Please see instructor as soon as possible to discuss possible classroom accommodations.

<u>NAME</u>	<u>CAS No.</u>
Acetic acid	64-19-7
Acetone	67-64-1
Ascorbic acid	50-81-7
Barium chloride	10361-37-2
Calcium chloride	10043-52-4
Cupric nitrate	19004-19-4
Ethanol	64-17-5
Hexanes	110-54-3
Hydrochloric acid	7647-01-0
Glycerol	56-81-5
Iodine	7553-56-2
Lithium chloride	7447-41-8
Magnesium	7439-95-4
Potassium bromide	7758-02-3
Potassium carbonate	584-08-7
Potassium chlorate	3811-04-9
Potassium chloride	7447-40-7
Potassium hydrogen carbonate	298-14-6
Potassium nitrate	7757-79-1
Potassium oxalate	6487-48-5
Potassium permanganate	7722-64-7
Potassium phosphate	7758-53-2
Potassium sulfate	7778-80-5
Silver nitrate	7761-88-8
Sodium bicarbonate	144-55-8
Sodium carbonate	497-19-8
Sodium chloride	7647-14-5
Sodium hydroxide	1310-73-2
Sodium metal	7440-23-5
Strontium chloride	10025-70-4
Zinc	7440-66-6

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0.2: Emergency Procedures

Emergency Procedures

Immediately alert the instructor to ANY fire or accident.

Fire Alarm

In case of a fire alarm, turn off all burners, water, and electrical devices. Collect any valuables (calculators, purses, etc.) and exit the building by assigned routes. **DO NOT RUN.** A copy of the evacuation route for your laboratory is posted on the laboratory bulletin board. Be familiar with the exit routes. Go to your assigned assembly area outside the building and remain there until the instructor has determined that all students are accounted for. Do not return to the building until given permission by your instructor or a campus security officer.

Injury

Immediately notify the instructor of ANY injury. First aid supplies are available in the laboratory. Minor injuries or burns may be treated by the laboratory instructor. Students with more serious injuries will be escorted to the student health center or a local hospital emergency room.

Medical Emergency Room

All medical emergencies (life-threatening situations, chest pain, excessive bleeding, eye injuries, head injuries, ingestion or inhalation of toxic substance, seizures, serious allergic reaction, suspected fracture) should be reported immediately to Wake EMS by dialing 9-911 from a campus phone and then Campus Police contacted by dialing 8888 from a campus phone or (919)-760-8888 from any other phone.

Emergency Information

Meredith College guidelines for emergency situations can be found online at the link below. It is your responsibility to read this brochure and be familiar with the information.

<https://www.meredith.edu/emergency-planning/emergency-procedures/>

Emergency Phone Numbers

Campus Police 8888

Fire, Police, Rescue 9-911

Campus Health Services 8535

Environmental Emergency 800-424-8802

Campus Maintenance 8560 (to shut off gas, water, or electricity if necessary)

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0.3: Safety Dos and Don'ts

SAFETY REGULATIONS FOR CHEMISTRY LABORATORIES

DO THE FOLLOWING whenever you are in the laboratory:

1. Wear approved eye protection at all times. If you get a chemical in your eye, wash the eye with flowing water from the eye wash station for at least 15 minutes. Forcibly hold the eye open for effective washing.
2. Dress appropriately for lab. Wear clothing that will protect your skin. Pants, skirts, or dresses extend to the ankle with no skin showing. Sleeveless shirts and bare midriffs are not permitted. Shoes must cover the entire foot. Sandals, open shoes, or crocs are not permitted. Confine long hair.
3. Place backpacks in an area where they will be out of the way and do not present a tripping hazard or block access to emergency equipment (safety showers, fire extinguishers) or exits.
4. In case of fire or accident, call the instructor at once. Know the location of the fire extinguishers, fire blanket, eyewash stations, safety shower, and first aid kit, and know how to use them. If your clothing catches fire, immediately DROP to the floor and ROLL over to smother the flames. Never use a fire extinguisher of any type on a person; instead use a fire blanket to smother the flames.
5. Immediately report any injury, no matter how minor, to the instructor.
6. If a fire alarm sounds, turn off any gas valve or water that is in use. Collect your valuables and calmly exit the building by the nearest exit. Meet in the assembly area designated by your instructor and remain there until the instructor has determined that everyone is out of the building.
7. Be extremely careful lighting or using a gas burner. Never leave a burner unattended.
8. Carefully read labels on reagent bottles twice before using. Using the wrong chemical may have serious consequences. Consider all chemicals to be hazardous. Protective gloves will be provided for those experiments that require them. Safety Data Sheets (SDS) are available for all chemicals.
9. Clean up any spilled materials immediately. Notify the instructor to obtain any special instructions for cleaning.
10. Avoid breathing any fumes. Use the snorkel hoods at the lab stations or the fume hoods for any experiments that will generate fumes.
11. Follow the procedures for disposal of chemical wastes as given in the laboratory manual or by the instructor.
12. If any chemical contacts your skin, immediately wash it off with plenty of running water. The safety shower should be used only if a large area of your body is exposed to a corrosive or toxic chemical. Immediately remove any contaminated clothing.
13. Clean up broken glassware immediately and dispose of the glass in a container marked "BROKEN GLASS ONLY." NEVER use glassware that is broken, cracked, or chipped.
14. When you are finished in the laboratory, thoroughly clean your work area with a sponge. Check to be certain that the gas and water are turned off and that all of your equipment is put away. Wash your hands thoroughly before leaving the laboratory.

DO NOT DO THE FOLLOWING when you are in the laboratory:

1. Do not eat, drink, smoke, or engage in horseplay.
2. Do not work in a laboratory unless an instructor is present. Never perform any unauthorized experiment.
3. Do not insert your spatula or pipet directly into a reagent bottle because it may result in contamination.
4. Do not allow papers to accumulate in your work area; they are a fire hazard.
5. Do not point a test tube containing a reacting mixture toward yourself or another person.
6. Do not add water to concentrated acids because the mixture may become extremely hot. Always add the acid slowly to water with constant stirring.
7. Do not remove chemicals from the laboratory.

8. Do not allow visitors.
9. Do not use cell phones.
10. Do not place calculators or computers where chemicals might be spilled on them.

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0.4: Hazard Communication

Communication of Chemical Hazards

The Hazard Communication Regulation of the Occupational Safety and Health Administration (OSHA) requires that anyone who might come in contact with chemicals be informed of potential hazards. Communication of chemical hazards is through labels and safety data sheets as described below.

The Fire Diamond

A quick visual representation of hazards is provided by the Health, Flammability, Instability (HFI) or “Fire” diamond developed by the National Fire Protection Association (NFPA). The diamond consists of four quadrants with numbers or special symbols representing the degree of certain hazards. The top three quadrants contain National Fire Protection Association (NFPA) codes (numbers from 0 to 4) indicating the degree of a particular risk.

4 – Extreme Risk

3 – Serious Risk

2 – Moderate Risk

1 – Slight Risk

0 – Minimal Risk

Sample of an HFI Diamond



Interpretation of HFI Diamond

Quadrant	Type of Hazard	Color	Example
left	health	blue	3 = severe health hazard
top	flammability	red	4 = extremely flammable
right	instability	yellow	2 = moderate reactivity
bottom	special	white	W = use no water

Hazard Pictograms

OSHA required hazard pictograms on labels consist of a black hazard symbol on a white background framed within a red border in the shape of a square set on a point. You should be familiar with these symbols.

Health Hazard  • Carcinogen • Mutagenicity • Reproductive Toxicity • Respiratory Sensitizer • Target Organ Toxicity • Aspiration Toxicity	Flame  • Flammables • Pyrophorics • Self-Heating • Emits Flammable Gas • Self-Reactives • Organic Peroxides	Exclamation Mark  • Irritant (skin and eye) • Skin Sensitizer • Acute Toxicity • Narcotic Effects • Respiratory Tract Irritant • Hazardous to Ozone Layer (Non-Mandatory)
Corrosion  • Skin Corrosion/Burns • Eye Damage • Corrosive to Metals	Exploding Bomb  • Explosives • Self-Reactives • Organic Peroxides	Gas Cylinder  • Gases Under Pressure
Flame Over Circle  • Oxidizers	Skull and Crossbones  • Acute Toxicity (fatal or toxic)	Environment  Not Mandatory • Aquatic Toxicity

For a more complete discussion of hazard pictograms, visit the OSHA Web site. www.osha.gov

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0.5: Safety Data Sheets

Safety Data Sheet (SDS)

Chemical manufacturers are required by law to provide a Safety Data Sheet (or SDS) for each chemical. Prior to 2015, this document was called a Material Safety Data Sheet or MSDS. Although your instructor or laboratory manual will always inform you of any hazards associated with the chemicals you will be using, you may view the SDS for any chemical you will use in the laboratory. You should be familiar with the location of the SDS file.

The format of a Safety Data Sheet is standardized to contain 16 sections, each providing a specific type of information.

Section	Section Heading	Information
1	Identification of the substance or mixture and of the supplier	
2	Hazards identification	
3	Composition/information on ingredients Substance/Mixture	The CAS (Chemical Abstract Service) number is a unique identifier and is especially important when accessing computer databases.
4	First aid measures	
5	Firefighting measures	
6	Accidental release measures	
7	Handling and storage	This section gives advice on proper storage of the chemical.
8	Exposure controls/personal protection	
9	Physical and chemical properties	
10	Stability and reactivity	These sections provide information about chemical incompatibilities and other chemical reaction dangers.
11	Toxicological information	Common abbreviations used in this section are listed below. Normally chemical exposure involves a chemical entering your body through inhalation (breathing), eye contact, skin absorption, or orally (ingestion). HMN: Human IVN: Intravenous (in the blood stream) MUS: Mouse ORL: Oral SKN: Skin TC _{Lo} : Toxic concentration low TD _{Lo} : Toxic dose low LC ₅₀ : Lethal Concentration 50; The concentration of a substance in air that can kill 50% of the test animals when administered as a single respiratory exposure LD ₅₀ : The lethal dose that will cause death in 50% of the test subjects.
12	Ecological information (<i>non mandatory</i>)	
13	Disposal considerations (<i>non mandatory</i>)	
14	Transport information (<i>non mandatory</i>)	
15	Regulatory information (<i>non mandatory</i>)	
16	Other information including information on preparation and revision of the SDS	

To search for a SDS for a chemical, go to <http://www.ilpi.com/msds/index.html>

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0.6: SDS Activity

Using the sample SDS for sodium bicarbonate (baking soda), find the following information.

https://www.fishersci.com/content/dam/fishersci/en_US/documents/programs/education/regulatory-documents/sds/chemicals/chemicals-s/S25533.pdf

CAS Number _____ Chemical Formula _____

First Aid (Eye Contact) _____

Eye Protection _____

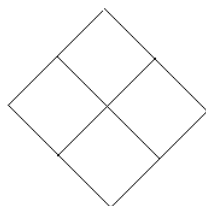
NFPA Ratings Health _____ Fire _____ Instability _____ Special _____

Signal Word _____

Hazard Statements _____

Hazard Pictogram _____

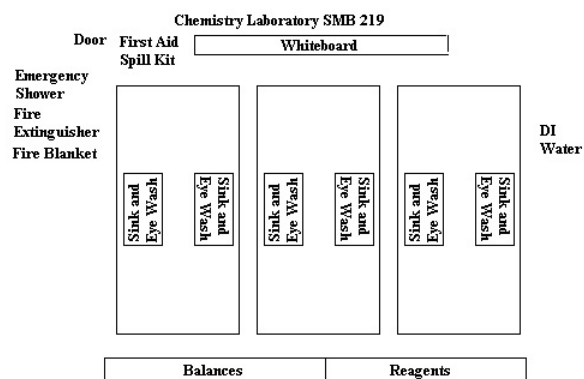
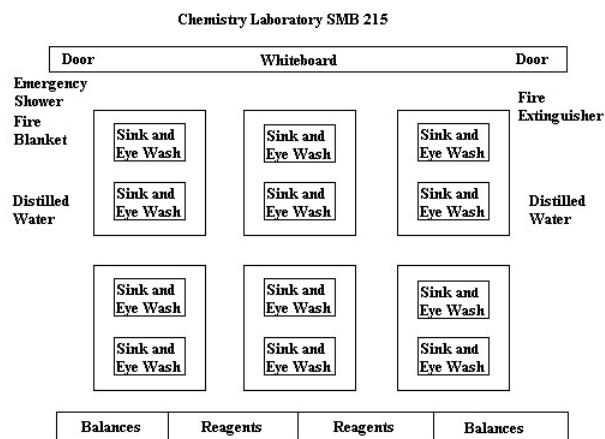
Complete the fire diamond for sodium bicarbonate.



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0.7: Lab Layout

Laboratory Layout



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0.8: Check in/out

Check in

Remove all the equipment from your lab drawer and cupboard. Go through the list of equipment and identify each piece. If the glassware is clean, return it to the drawer or to the cupboard.

If any glassware is chipped, cracked, or broken, then discard it in the container marked **BROKEN GLASS ONLY**.

If any of your glassware or equipment is missing or broken, write the name of the item on a piece of paper with your name. The instructor or a laboratory assistant will get the missing item for you.

If you have extra items, place them in the tray designated for excess equipment.

If any piece of glassware is dirty, clean it with the soap solution in the spray bottle at your bench. Dry the outside with a paper towel, invert on a paper towel and allow most of the water inside to drain out before returning the glassware to the drawer or

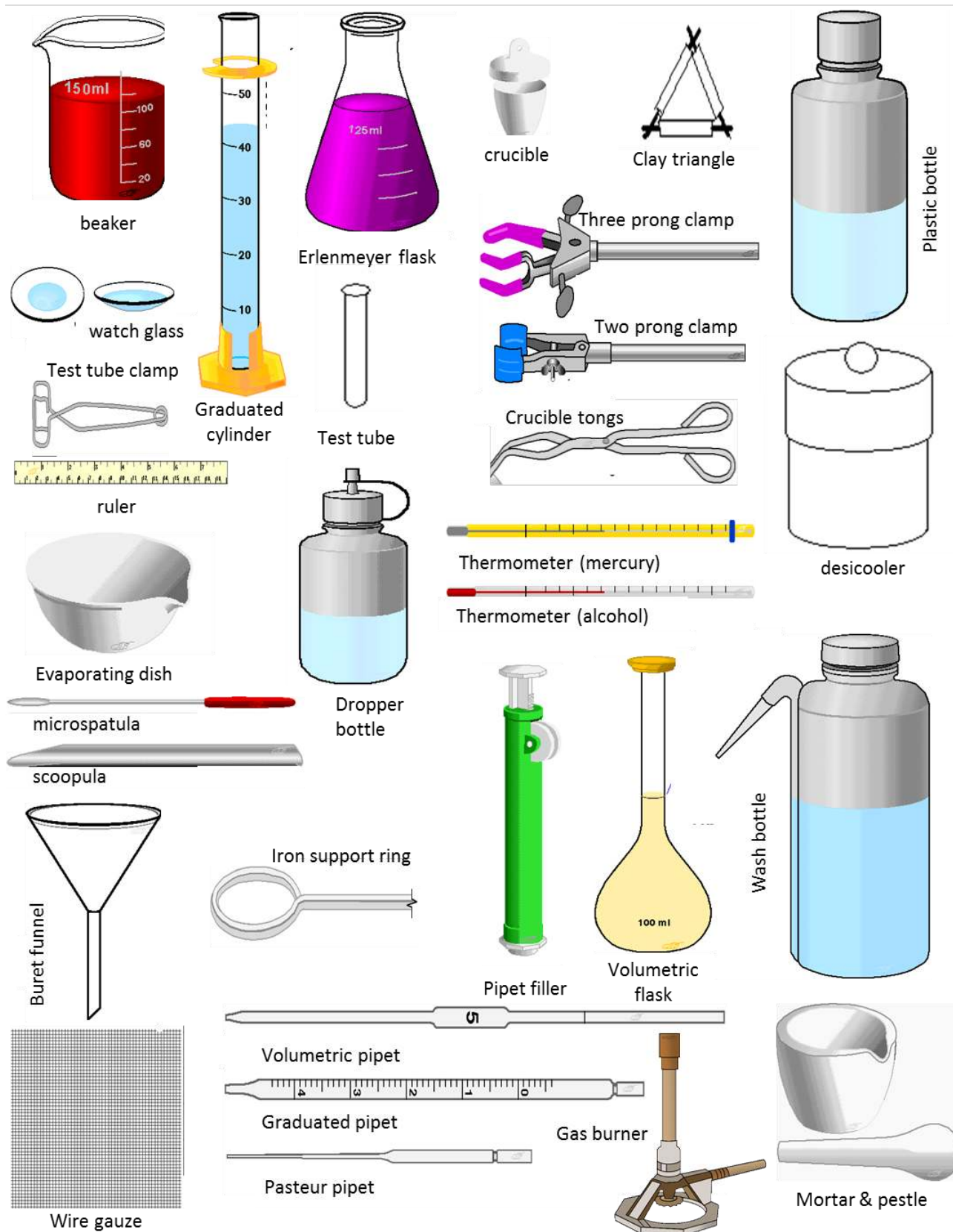
cupboard.

Quantity	Item/Quantity	Size/Description
Equipment in Drawer		
6	Beakers	2 – 50 mL 2 – 100 mL 1 – 150 mL 1 – 250 mL
1	Buret Card	
1	Evaporating Dish	
1	Flask, Erlenmeyer	50 mL
3	Flasks, Erlenmeyer	250 mL
1	Graduated Cylinder	10 mL
1	Graduated Cylinder	50 or 100 mL
2	Litmus paper	1 red, 1 blue
3	Pipets	Small
2	Rubber bulbs	
1	Ruler	
1	Spatula/Scoopula	
5	Stirring rods	4 small, 1 large
1	Test tube clamp	
1	Test tube rack	
17	Test Tubes	2 – 150 x 18 mm 3 – 100 x 13 mm 12 – 75 x 10 mm
1	Watch glass	
<u>Community Equipment in Cupboard</u>		
2	Bottles	Plastic, with lids – 1 L
2	Brushes	1 small, 1 large
1	Buret Funnel	
1	Buret holder	
1	Burner	with rubber tubing
1	Clay triangle	
1	Dessicooler	
1	Flask clamp	
2	Flasks, Volumetric	100 mL
1	Iron ring	

1	Thermometer	
1	Tongs	
1	Wash bottle	
2	Wire gauze	

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0.9: Lab equipment



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CHAPTER OVERVIEW

Experiment 1 - Laboratory Techniques

[1.1: Pre-lab](#)

[1.2: Procedure](#)

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1.1: Pre-lab

Analytical Balance – Print the section of the *Laboratory Handbook* entitled “Analytical Balance (AB 104) (Operating Instructions).” Go to the following Web site

<http://academics.hamilton.edu/chemistry-lab-site/analytical-balance>

and review the information about using an analytical balance. The analytical balance demonstrated at this site is not exactly the same as the ones in our laboratories; however, it is very similar.

Buret – Print the section of the *Laboratory Handbook* entitled “Burets.” Go to the following Web site:

<https://www.youtube.com/watch?v=48do-3bzq3U>

and review the information about using a buret. You should pay special attention to how to condition the buret, fill it with solution and check for air bubbles and leaks.

Read the volume of the liquid in each of the burets shown in the Buret section of the *Laboratory Handbook*. Record your results below.

First Buret _____ Second Buret _____

Pipet – Print the section of the *Laboratory Handbook* entitled “Pipets.” Go to the Web site <https://www.youtube.com/watch?v=2WZ6sREzCsc> and review the technique for transferring a known volume of liquid using a volumetric (transfer) pipet.

Save the *Laboratory Handbook* sections for future reference.

Chemical Calculations – Use your textbook to review the calculations for determining molar mass, mole to mass conversions, mass to mole conversions, and solution concentration using molarity.

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1.2: Procedure

During this laboratory period you will have an opportunity to practice several important laboratory skills. You may perform the following procedures in any order.

I. Determining Mass Using An Analytical Balance – Determining mass is one of the most common and most important laboratory operations. Accurate masses are crucial to good results. You have previously used balances that determine mass to the nearest 0.01 or 0.001 g. For the most precise analytical work or for very small samples, you will need to use an analytical balance that can determine mass to the nearest 0.0001 g.

Obtain a set of unknown masses. DO NOT TOUCH the bottle with bare fingers. Use a paper towel or tongs. Record the sample number in the space below. Remove the cap from the bottle. Determine the mass of the bottle and contents, including units. Using forceps, remove the mass from the bottle. Place the mass on a clean paper towel. Re-weigh the bottle and record the mass. Using forceps, replace the masses in the bottle.

Sample Number _____

Mass of bottle including all unknowns _____

Mass of bottle minus first unknown _____

Mass of first unknown _____

II. Reading A Buret – There are several burets set up in the laboratory for you to practice reading. Read each practice buret and compare your reading to the reading recorded on the card next to it. You should be within ± 0.01 mL of the recorded reading.

III. Transferring Known Volume Using A Volumetric Pipet - Obtain ~50 mL of DI water in a clean beaker. Determine the mass of another small, clean, dry beaker using a top-loading balance. Use a 5.00 mL volumetric pipet to transfer 5.00 mL of the DI water to the small, dry beaker. Weigh the small beaker and the water. Determine the mass of the water, then using its density determine the volume of water transferred. Finally, determine the percent error. Repeat this process until you can obtain accuracy (percent error) of ~1%.

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
Mass of empty beaker					
Mass of beaker plus 5.00 mL water					
Mass of water					
Density water					
Calculated volume of water					
Volume % error					

IV. Chemical Calculations: Answers should have appropriate units and significant figures.

4a. Determine the molar mass of ammonium phosphate, $(\text{NH}_4)_3\text{PO}_4$. Express your answer to two decimal places.

4b. How many grams of ammonium phosphate are there in a 0.255 mol sample?

4c. 42.87 g of ammonium phosphate represents how many moles?

4d. What is the molarity of a solution prepared by dissolving 6.22 g of ammonium phosphate in enough water to make 500.0 mL of solution?

4e. How many grams of ammonium phosphate are required to prepare 250.0 mL of 0.300 M ammonium phosphate?

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CHAPTER OVERVIEW

Experiment 2 - Exploring Solubility and Net Ionic Equations

[2.1: Background](#)

[2.2: Pre-lab](#)

[2.3: Safety](#)

[2.4: Procedure](#)

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2.1: Background

Solubility, the ability of one substance to dissolve in another, plays an important role in a variety of everyday processes. We are able to inject vaccines into the body because of their solubility and gain vitamins from our food because of their solubility. Insolubility is also a crucial characteristic. Our houses do not dissolve in rain because of their insolubility. Kidney stones become a problem because their lack of solubility causes pain and can lead to infection.

We have learned previously that solubility can be determined by the similarity of the intermolecular forces (IMFs) between a solvent and a solute; the more similar the IMFs are, the more soluble the compounds are. When a substance is soluble, you can determine its concentration using molarity ($M=n/V$). When a substance is insoluble or a precipitate, its concentration is 0 M as nothing is dissolved.

However, solubility itself does not tell the full story. Both table salt (sodium chloride) and table sugar (sucrose) are soluble, but there is an important difference between the two solutions. The salt solution is a good conductor of electricity, whereas the sugar solution is not. The conductivity of the salt solution is due to the presence of ions. The salt dissociates into ions when it dissolves; the sugar molecules do not dissociate. Substances that dissociate into ions in solution are called electrolytes. Substances that do not form ions in solution are called nonelectrolytes.

Question

Why are their differences in behavior between salt and sugar?

Answer

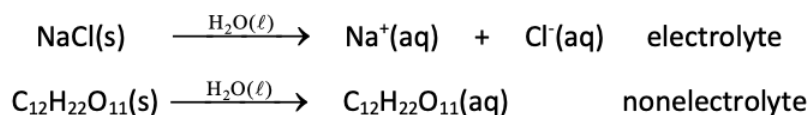
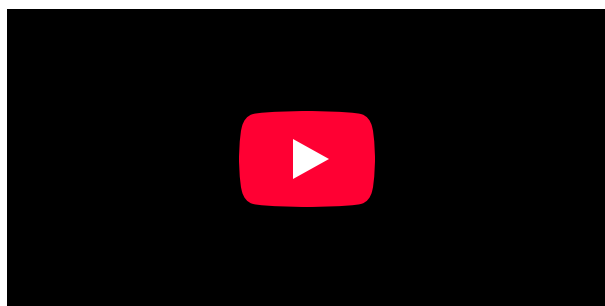
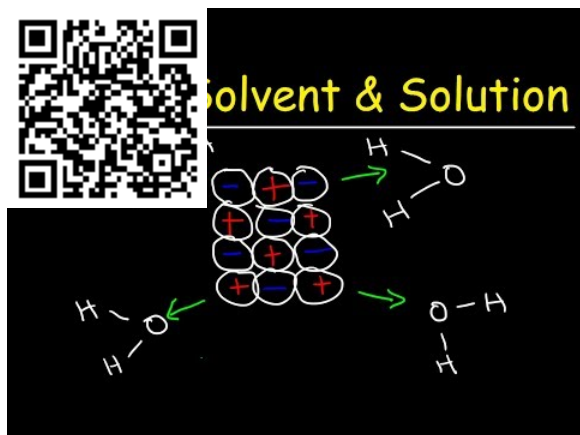


Figure : Salt and Sugar Net Ionic Equations.

Sugar is a covalent compound and thus its bonding is not disrupted by the presence of water molecules. Salt, on the other hand, is an ionic compound. When an ionic compound dissolves in water, the anions and cations that make up the crystal lattice are pulled apart by the attraction to water molecules. A water molecule is polar, that is, although it is a neutral molecule, the electrical charge is not evenly distributed. The oxygen end of a water molecule has a partial negative charge, and the hydrogen end has a partial positive charge.

Go to the Web site below to view an animation of salt dissolving in water. Be sure the sound is turned on so that you can hear the narration. Although only a few water molecules are shown for clarity, the crystal of sodium chloride is surrounded completely by water molecules. For your pre-lab, **watch the first 7 minutes of the following video and answer the following questions.**





When dissolution occurs, that is when the ionic compound solubilizes, the ionic bond in the compound is broken. The energy required to break this bond is known as lattice energy (Figure 1.1.2) and is defined by the electrostatic attraction between the cation and anion in the compound.

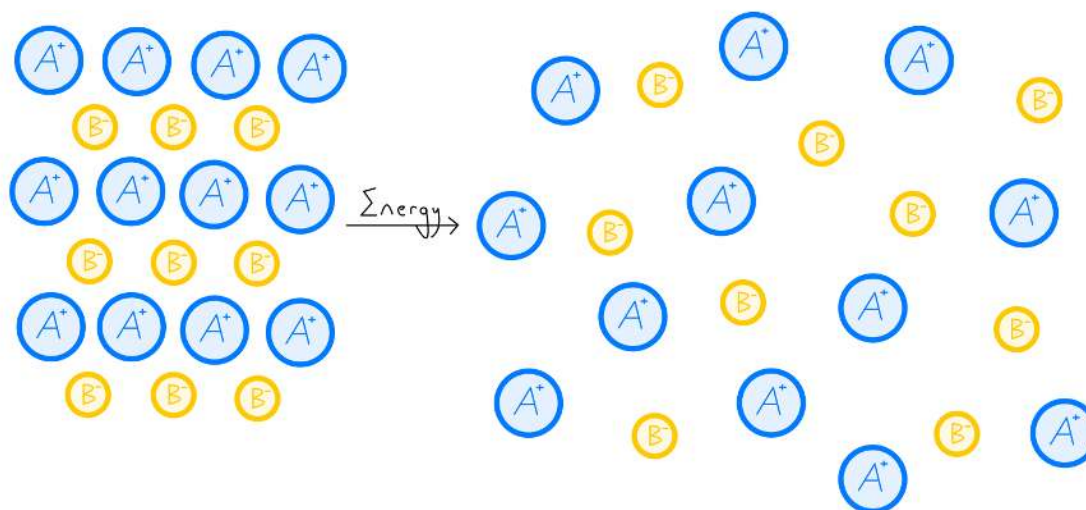


Figure 1.1.2: Diagram demonstrating lattice energy.

Electrostatic energy is defined by

$$F = k \frac{Q_1 Q_2}{r^2}$$

Where Q_1 and Q_2 are the charges on the cation and anion and r is the ionic bond distance.

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2.2: Pre-lab

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2.3: Safety

DISPOSAL and CLEANUP: Use a disposable pipette to withdraw all of the solution and solid material from the glassware containing barium. Soluble barium salts are highly toxic. Dispose of this material in the container labeled for barium wastes. Thoroughly rinse the contents of the remaining glassware down the sink drain with large amounts of water.

Chemicals	Hazards	Pictograms	Required PPE
KNO_3	May intensify fire		Glasses/Goggles Gloves
K_2SO_4	n/a	n/a	Glasses/Goggles Gloves
K_2CO_3	Skin irritation Eye irritation Respiratory irritation		Glasses/Goggles Gloves
K_3PO_4	Serious eye damage Respiratory irritation	 	Glasses/Goggles Gloves
NaCl	n/a	n/a	Glasses/Goggles Gloves
CaCl_2	Eye irritation		Glasses/Goggles Gloves
LiCl	Harmful if swallowed Skin irritation Eye irritation Respiratory irritation		Glasses/Goggles Gloves
BaCl_2	Toxic if swallowed Causes serious eye irritation Harmful if inhaled	 	Glasses/Goggles Gloves
CaBr_2	Causes serious eye damage		Glasses/Goggles Gloves
MgSO_4	n/a	n/a	Glasses/Goggles Gloves
MgO	n/a	n/a	Glasses/Goggles Gloves
NH_4Cl	Harmful if swallowed Causes serious eye irritation		Glasses/Goggles Gloves

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2.4: Procedure

Part 1: Compound Solubility

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CHAPTER OVERVIEW

Experiment 3 - Colligative Properites

[3.1: Background](#)

[3.2: Pre-lab](#)

[3.3: Safety](#)

[3.4: Procedure](#)

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3.1: Background

Colligative properties are defined as solution properties that depend solely on the concentration of the solute particles, not their identity. These properties include changes in boiling point, freezing point, vapor pressure and osmotic pressure. For instance, magnesium chloride and calcium chloride both lower the freezing point of water to -20°C despite having different chemical identities.

We can define the change in colligative properties of a solution based on the following equations with constants in reference to the solvent:

$$T_b = imK_b$$

$$T_f = imK_b$$

$$P_{\text{solution}} = iX_{\text{solvent}}P_{\text{solvent}}^{\circ}$$

$$\pi = iMRT$$

We can use colligative property changes as a tool to help us determine the identity of a compound, such as whether it is an electrolyte or not, or even the number of ions produced when a solute is dissolved in a solvent. You will be using colligative properties, along with other tests performed in this class and general chemistry I, to determine the identity of an unknown substance found at a crime scene.

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3.2: Pre-lab

What is the relationship between vapor pressure and solution concentration? **Explain** why on a sub-microscopic level.

As vapor pressure increases, what happens to boiling point? **Explain**.

What are the other relationships (ie what changes are observed with the addition of a non volatile salt) that are associated with colligative properties?

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3.3: Safety

DISPOSAL and CLEANUP: Everything can go down the drain with lots of water.

Compound	Hazards	Pictograms	Required PPE
NaCl	n/a	n/a	Glasses/Goggles Gloves
LiCl	Harmful if swallowed Skin irritation Eye irritation Respiratory irritation		Glasses/Goggles Gloves
K ₂ CO ₃	Skin irritation Eye Irritation Respiratory irritation		Glasses/Goggles Gloves
CH ₃ OOH	Flammable liquid and vapor Causes severe burns and eye damage	 	Glasses/Goggles Gloves
Unknown	Harmful if swallowed Causes serious eye damage		Glasses/Goggles Gloves

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3.4: Procedure

Part I: Exploring Colligative Properties

We are going to be looking at how boiling points change with solute changes for a variety of different solutes.

For each compound below, make 25 mL of 4 M aqueous solution in a 50 mL Erlenmeyer flask. Be sure to show your work for how much solute you added.

LiCl

NaCl

K₂CO₃

CH₃COOH (10 M solution)

Determine the boiling points of DI water and all of these solutions under the fume hood.

Describe what happened to the boiling points of the solutions compared to the boiling point of water. Remember we must use experimental data for all boiling points.

Rank the boiling points of the solutions from lowest to highest. Explain the chemical rationale behind your observations.

Part II: Unknown Substance Identification

The Raleigh police found an unknown powder at the scene of a local robbery and are hoping to identify it to get some leads on the suspect. Based on the powder's physical appearance, the police have narrowed it down to this list of options. Next to each option is listed the use of each chemical.

NaCl (table salt)	Na ₂ SO ₄ (laundry detergent)	CaSO ₄ (dentistry mold)	K ₃ PO ₄ (cooking)	Na ₃ PO ₄ (cleaner)
CaCl ₂ (deicer)	K ₂ SO ₄ (fertilizer)	MgSO ₄ (laxative)	C ₈ H ₁₀ N ₄ O ₂ (caffeine)	FeSO ₄ (treats anemia)
C ₆ H ₁₂ O ₆ (sugar)	CuCl ₂ (dyes)	Mg(OH) ₂ (flame retardant)	C ₉ H ₈ O ₄ (aspirin)	NiSO ₄ (battery production)
FeCl ₂ (water treatment)	FeCl ₃ (copper etching)	KCl (explosives)	NiCl ₂ (glass coloring)	C ₁₀ H ₁₃ NO ₂ (phenacetin: cocaine cutting agent)

The police have given you the powder in a 4 M solution. Using tests that you have used in this course and general chemistry I lab, determine the identity of the powder. *It might be helpful to think about ways to determine bond type, cation/anion identity, if applicable, and the number of cations and anions in the compound, if applicable.*

Describe your proposed procedure here.

Write any observations or results from your procedure. If you deviated from your procedure, be sure to note that as well.

Determine the identity of the substance. Justify your answer using your experimental data.

Explain why knowing the identity of the substance be valuable for law enforcement.

Write a **brief** (~one paragraph) report to the law enforcement agency with the results of your tests and the identification of the powder. It is important for the agency to reference why each test was used and how it led to your conclusion.

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CHAPTER OVERVIEW

Experiment 4 - Molecular Absorption Spectroscopy

[4.1: Background](#)

[4.2: Pre-lab](#)

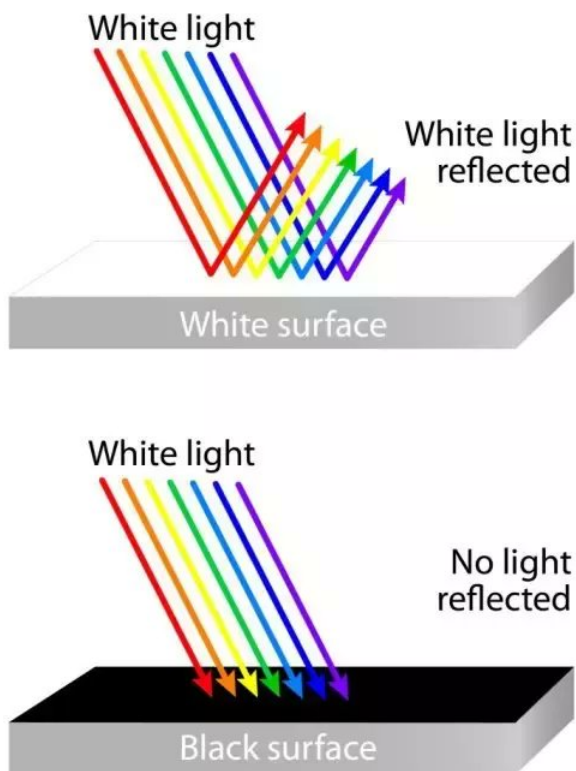
[4.3: Safety](#)

[4.4: Procedure](#)

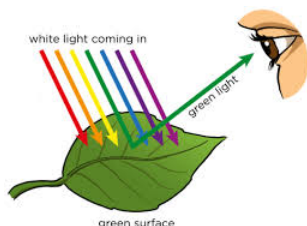
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4.1: Background

Chemists can identify a substance by the color (wavelength) of light it emits. It is also possible to identify a substance by the color (wavelength) of light it absorbs. White light consists of all the wavelengths of electromagnetic radiation from about 400 nm to 700 nm. When white (colorless) light enters a sample and all of the colors of light are strongly absorbed the sample appears black. If a sample absorbs none of the colors of light, then it appears white or colorless.



A more interesting situation occurs when a sample absorbs some of the colors of light. The remaining colors pass through the solution. The color we observe for the solution is the sum of these remaining colors that strike your eyes.



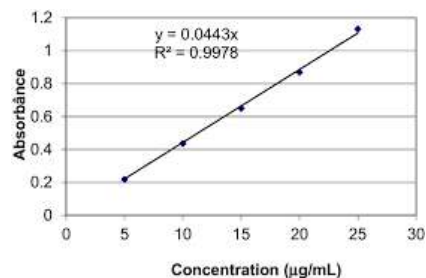
The absorbance is directly proportional to the concentration, that is, the greater the concentration of a solution, the more light that is absorbed. Additionally, the path length, or the size of the sample, can affect the amount of light or “photons” absorbed. The greater the path length, the more molecules are present to absorb photons, thus, the greater the absorbance value. The relationship is known as Beer’s law or

$$A = kc$$

where c is the concentration (molarity) and k is a constant that is dependent on the wavelength. The absorbance is related to the intensity of the color of a solution. If we compare two solutions of the same substance and the color of one appears to be more intense than the color of the other, then we can reasonably assume that the darker solution is absorbing more light (has a higher absorbance) and is the more concentrated solution.

In determining the concentration of a solution, a calibration curve is normally used. Solutions of known concentration are prepared, and their absorbances are measured and plotted on a graph of absorbance versus concentration. The absorbance of the unknown solution is then measured and its concentration read from the calibration curve.

Calibration Curve:



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4.2: Pre-lab

1. You have in your lab: 1.5 M NaNO_3 , 100.0 mL volumetric flask, 1.000 L volumetric flask, a graduated cylinder, and a balance. Write a detailed set of instructions on how to prepare 100.0 mL of 0.100 M NaNO_3 .
 2. If a sample absorbs none of the colors of light, what color does it appear? Explain.
 3. If a solution has a low concentration, does it absorb more or less photons (amount of light) than a solution with higher concentration? Explain.
-

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4.3: Safety

Chemicals	Hazards	Pictograms	Required PPE
Red Dye #40	NA	NA	Glasses/goggles Gloves

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4.4: Procedure

In this experiment, you will prepare several solutions of a food dye of known molarity. By measuring the absorbance of each of these solutions you will be able to prepare a calibration curve. Finally you will measure the absorbance of an unknown solution and determine its concentration using your calibration curve. For this experiment you will need the following items:

- 4 medium test tubes
- 2 10 mL graduated cylinders
- 1 small stirring rod
- 6 cuvettes
- 1 100 mL beaker
- parafilm

To obtain accurate results in this experiment, you must use extreme care in measuring ALL volumes!!

1. Thoroughly clean and rinse with **deionized water** all of the glassware for this experiment. Do not wash cuvettes with a brush. Invert the 150 x 18 mm test tubes on a paper towel and allow them to drain dry. Label the test tubes A, B, C, and D. These test tubes must be **completely** dry.
2. Obtain approximately 15 mL of the standard solution in the clean, dry test tube A. Rinse one of your 10 mL graduated cylinders with a small portion of this solution. Use this cylinder for all solution transfers. Reserve your second graduated cylinder for deionized water only.
3. Transfer exactly 7.50 mL of solution A into test tube B. Use a disposable pipet to dispense the last couple of drops to obtain an exact measurement. Using your second graduated cylinder, add exactly 7.50 mL of deionized water to test tube B. Use a disposable pipet (reserved for deionized water) to obtain an exact measurement. Mix the solution thoroughly with a small, clean, dry stirring rod.
4. Rinse your solution pipet with a small sample of solution B and then transfer exactly 7.50 mL of this solution into test tube C. Also transfer exactly 7.50 mL of deionized water into test tube C. Mix thoroughly with a clean, dry stirring rod.
5. Rinse your solution cylinder with a small portion of solution C and then transfer exactly 7.50 mL of this solution into test tube D. Also transfer exactly 7.50 mL of deionized water into test tube D. Mix thoroughly with a clean, dry stirring rod.
6. You now have four solutions, each exactly half the concentration of the solution from which it was prepared. Record the concentration of each solution on the data sheet. You may use the formula: $M_{\text{concentrated}} \times V_{\text{concentrated}} = M_{\text{diluted}} \times V_{\text{diluted}}$.
7. The blank may already be prepared for you, if not, fill one of your cuvettes two-thirds full with deionized water. Wipe away any excess water with a Kimwipe.
8. Rinse another cuvette with a small portion of solution A and then fill this cuvette two-thirds full with solution A. Wipe away any excess solution with a Kimwipe. With the help of your TA, determine the absorbance of this solution and record the result on the data sheet.
9. Empty the cuvette containing solution A and repeat step 8 with each of the remaining solutions.
10. Prepare a calibration curve by plotting the absorbance (y-axis) versus the concentration (x-axis). Use Excel to create a trend line with its equation.
11. Using a clean, dry test tube, obtain a sample of the unknown solution. Make a visual comparison of your unknown with the solutions A, B, C, and D. Can you roughly estimate the concentration of your unknown by comparing it to your standard solutions?
12. Repeat step 8 using your unknown solution, no dilution necessary.
13. Using your calibration curve, determine the concentration of your unknown.
14. Thoroughly clean and rinse the cuvettes. Do not use a brush.

DISPOSAL: All solutions may be rinsed down the sink drain with plenty of water.

Standard Solution _____

Solution Concentration Absorbance

A _____

B _____

C _____

D _____

unknown _____ unknown ID: _____

Attach your calibration curve to the assignment (with appropriate titles) on Brightspace.

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CHAPTER OVERVIEW

Experiment 5 - Heat of Reaction

[5.1: Background](#)

[5.2: Pre-lab](#)

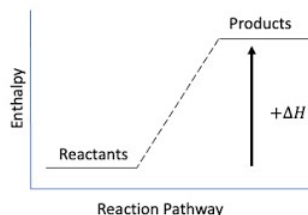
[5.3: Safety](#)

[5.4: Procedure](#)

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5.1: Background

All chemical reactions involve either the evolution or absorption of energy, usually in the form of heat. Heat is released in **exothermic** reactions and absorbed in **endothermic** reactions. The change in energy can be shown in an energy diagram. An energy diagram for the combustion of solid carbon is shown below.



Since this reaction is **endothermic**, the system absorbs energy from the surroundings and the energy of the products that remain is greater than the energy of the original reactants. ΔE for an endothermic reaction is positive indicating that the energy of the system has increased. For exothermic reactions, energy is released by the system, and the energy of the products is less than the energy of the reactants. ΔE is negative indicating that the energy of the system has decreased.

The energy change that occurs during a chemical reaction can be determined by measuring the temperature change that occurs when the reaction is carried out in an insulated container known as a **calorimeter**. The amount of heat involved depends upon the conditions under which the reaction is conducted. For example, most chemical reactions are carried out at constant pressure, usually atmospheric pressure. The heat evolved or absorbed at constant pressure is referred to as the **enthalpy change**, ΔH .

The temperature change experienced by a substance depends upon its **heat capacity**, C , which is defined by the equation

$$C = \frac{Q}{T_f - T_i} = \frac{Q}{\Delta T} \quad (1)$$

where Q is the quantity of heat added to the substance, T_i is the initial temperature of the substance, T_f is the final temperature of the substance, and ΔT is the change in temperature of the substance. Note that heat capacity and the temperature change are inversely related; that is, the greater the heat capacity, the smaller the temperature change for a given quantity of heat.

Another quantity related to heat capacity is the **specific heat capacity**, c . Some books use the symbol, C_s , for specific heat capacity. The specific heat capacity is the heat capacity per unit mass, m , of a substance and is defined as

$$c = \frac{C}{m} = \frac{Q}{m \Delta T} \quad (2)$$

The specific heat capacity of water is $4.184 \text{ J/}^\circ\text{Cg}$, which means that 4.184 J of heat are required to increase the temperature of 1 gram of water by 1°C . The heat capacity of aluminum is $0.89 \text{ J/}^\circ\text{Cg}$, thus only 0.89 J of heat are needed to raise the temperature of 1 gram of aluminum by 1°C .

To determine the quantity of heat required to increase the temperature of 10.0 grams of water from 25°C to 35°C , rearrange equation (2) to the following form.

$$Q = m c \Delta T \quad (3)$$

$$Q = (10.0 \text{ g})(4.184)(10^\circ\text{C}) = 4.2 \times 10^2 \text{ J}$$

Note that the temperature change has only two significant figures; therefore, the final answer is limited to only two significant figures.

In this experiment you will measure the heat involved in the reaction of a strong acid and a strong base. This enthalpy change is called the **heat of neutralization**. The heat given off in this reaction is mainly absorbed by the resulting solution; however, a small amount of heat is absorbed by the calorimeter.

heat given off by reaction = heat absorbed by solution + heat absorbed by calorimeter cup

$$\text{or } \Delta H = Q_{\text{solution}} + Q_{\text{calorimeter}}$$

$$\Delta H = m_{\text{solution}} \times C_{\text{solution}} \times \Delta T + m_{\text{calorimeter cup}} \times C_{\text{calorimeter cup}} \times \Delta T$$

The quantity ($m_{\text{calorimeter cup}} \times C_{\text{calorimeter cup}}$) is the heat capacity of the calorimeter cup, $C_{\text{calorimeter cup}}$, and is known as the **calorimeter constant**. The calorimeter constant is determined by measuring the temperature change for a process for which the quantity of heat is accurately known. For the styrofoam cups used for the calorimeter in this experiment, the calorimeter constant is 20. J/°C, that is, for each 1 °C increase in temperature the calorimeter absorbs 20. J of heat.

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5.2: Pre-lab

1. Calculate the heat capacity of a calorimeter using the data below. Show your calculations and write your final answer in the space provided below.

$$Q = 100. \text{ kJ} \underline{\hspace{2cm}}$$

$$T_{\text{initial}} = 21.0 \text{ }^{\circ}\text{C} \underline{\hspace{2cm}}$$

$$T_{\text{final}} = 27.0 \text{ }^{\circ}\text{C} \underline{\hspace{2cm}}$$

$$C_{\text{calorimeter}} = \underline{\hspace{2cm}}$$

2. Write the complete, balanced chemical equation and the net ionic equation for the neutralization of hydrochloric acid and sodium hydroxide.

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5.3: Safety

Chemicals	Hazards	Pictograms	Required PPE
1.00 M HCl	May be corrosive to metals Causes severe skin burns and eye damage Respiratory irritation		Glasses/Goggles Gloves
1.00 M NaOH	May be corrosive to metals Causes severe skin burns and eye damage Respiratory irritation		Glasses/Goggles Gloves

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5.4: Procedure

Construct a simple calorimeter by nesting two styrofoam cups together. Make a hole in the cardboard lid just large enough to insert the thermometer. Thoroughly rinse and dry the calorimeter. Place the calorimeter on a piece of cardboard to insulate it from the laboratory bench.

Obtain approximately 110 mL of 1.00 M hydrochloric acid in a 250 mL beaker. Carefully measure 50.0 mL of the 1.00 M HCl solution and add it to the calorimeter. Put the lid on the calorimeter, but do not insert the thermometer. Obtain approximately 110 mL of 1.00 M sodium hydroxide in a 250 mL beaker. Carefully measure 50.0 mL of the 1.00 M base solution in a clean, dry graduated cylinder. Measure the temperature of the base solution. Rinse and dry the thermometer and then measure the temperature of the acid solution. Accurate temperature measurements are crucial for good results in this experiment.

Lift the calorimeter lid and rapidly add all the base solution. Avoid splattering the solution. Insert the thermometer into the lid and gently stir the solution. Be sure the bulb of the thermometer is completely immersed in the solution. Watch the temperature rise on the thermometer and record the maximum temperature. Do not read the thermometer before the temperature stops rising. As the temperature of the mixture rises, heat from the reaction is lost to the calorimeter cup and its surroundings; therefore, the maximum observed temperature change is somewhat less than that actually attained by the reaction mixture. You will not need to correct for such heat losses in this experiment. Empty, rinse, and dry the calorimeter.

Repeat the above procedure for a second trial and average the results of the two trials. For computational purposes you may assume the specific heat capacity and density of your solution are approximately that of water, or 4.184 J/g °C and 1.00 g/mL, respectively.

DISPOSAL and CLEANUP: All solutions can be washed down the sink drain with plenty of water. Thoroughly rinse the calorimeter and dry with a paper towel.

DATA

Write the complete, balanced chemical equation for the reaction that occurs.

Write the net ionic equation for this reaction.

Enthalpy of Neutralization Trial 1 Trial 2

1. Molarity of HCl _____ M
2. Molarity of NaOH _____ M
3. Volume of acid solution _____ mL
4. Volume of base solution _____ mL
5. Total volume after mixing _____ mL
6. Density of solution (see previous page) _____ g/mL
7. Total mass of solution _____ g
8. Temperature of acid solution (T_A) _____ °C
9. Temperature of base solution (T_B) _____ °C
10. Average temperature before mixing _____ °C
11. Maximum temperature after mixing _____ °C
12. $\Delta T = T_{\text{after mixing}} - T_{\text{before mixing}}$ _____ °C
13. Heat gained by calorimeter cup = $C_{\text{cal cup}} \times \Delta T$ _____ J
($C_{\text{cal cup}} = 20. \text{ J/}^\circ\text{C}$)
14. Heat gained by solution = $m_{\text{soln}} \times c_{\text{soln}} \times \Delta T$ _____ J
15. Total heat of the surroundings = $Q_{\text{cal cup}} + Q_{\text{soln}}$ _____ J
16. Heat of the reaction _____ J
17. Moles of acid used _____ mol HCl
18. Moles of base used _____ mol NaOH
19. Moles of H_2O produced _____ mol H_2O
20. Molar enthalpy of reaction = $\Delta H = \text{total heat/mol H}_2\text{O}$ _____ kJ/mole*
21. ΔH average _____ kJ/mole (include sign)

DISCUSSION

Is this reaction endothermic or exothermic? Draw an energy diagram for the reaction.

What measured (not calculated or given) quantity limits the number of significant figures in the molar enthalpy?

Given the accepted value of the molar enthalpy for this reaction is -55.84 kJ/mol, what is the percent error for your experiment?

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CHAPTER OVERVIEW

Experiment 6 - Qualitative Inorganic Analysis

[6.1: Background](#)

[6.2: Pre-lab](#)

[6.3: Safety](#)

[6.4: Procedure](#)

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6.1: Background

Qualitative analysis involves determining the presence or absence of a particular substance. There are various reasons for including qualitative analysis in the general chemistry curriculum; development of careful lab techniques, critical observation skills, logical interpretation of results, and an increase in the understanding of ionic equilibria (weak electrolytes, solubility products, complex ions) and oxidation-reduction.

The table below lists 13 possible white, solid, unknown substances contained in this experiment. These compounds can be found in common household products and medications:

<u>COMPOUND</u>	<u>COMMON SOURCE</u>
-----------------	----------------------

Cornstarch ($C_6H_{10}O_5$) _n	Argo cornstarch
--	-----------------

Calcium Sulfate ($CaSO_4$)	Plaster of Paris
------------------------------	------------------

Calcium Carbonate ($CaCO_3$)	Chalk
--------------------------------	-------

Sodium Hydroxide (NaOH)	Draino (drain opener)
-------------------------	-----------------------

Sodium Carbonate (Na_2CO_3)	Arm & Hammer Washing Soda
---------------------------------	---------------------------

Potassium Carbonate (K_2CO_3)	Used in soaps and liquid shampoos
-----------------------------------	-----------------------------------

Magnesium Sulfate ($MgSO_4$)	Epsom Salts
--------------------------------	-------------

Boric Acid (H_3BO_3)	Boric Acid (dilute; used in eye drops)
--------------------------	--

Sodium Bicarbonate ($NaHCO_3$)	Arm & Hammer Baking Soda
----------------------------------	--------------------------

Sodium Chloride (NaCl)	Table Salt (uniodized)
------------------------	------------------------

Lithium Chloride (LiCl)	Found in antidepressants
-------------------------	--------------------------

Potassium Chloride (KCl)	Used in medicine as an electrolyte replenisher
--------------------------	--

(All three chlorides are also found in seawater)

Sucrose ($C_{12}H_{22}O_{11}$)	Common Table sugar
----------------------------------	--------------------

In this experiment, you will qualitatively analyze and identify the compounds listed above.

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6.2: Pre-lab

1. The first test to be performed in the experiment is a solubility test.

a. Describe how you will know if your sample is soluble or not.

b. Sodium chloride readily dissolves in water. Draw a molecular diagram showing sodium chloride dissolving in water and label the type of intermolecular forces present.

c. Sucrose readily dissolves in water. Draw a molecular diagram showing sucrose dissolving in water and label the type of intermolecular forces present.

2. Flame tests are used to distinguish between different metal cations. For the following cations, write the color of the flame when it is burned.

Cation	Flame Color
Na ⁺	
K ⁺	
Li ⁺	
B ³⁺	

3. Epsom salt, magnesium sulfate reacts with Drano, sodium hydroxide, to form a white precipitate. Write the balance molecular equation and include all phases.

4. Samples containing carbonate and bicarbonate salts will produce CO₂ gas bubbles when an acid is added to the solution. Write the balance molecular equation of the reaction between Na₂CO₃ and HCl. Include all phases.

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6.3: Safety

Chemicals	Pictogram	Required PPE
Sodium chloride	N/A	Nitrile gloves, glasses
Calcium sulfate	N/A	Nitrile gloves, glasses
Potassium chloride	N/A	Nitrile gloves, glasses
Calcium carbonate	N/A	Nitrile gloves, glasses
Corn starch	N/A	Nitrile gloves, glasses
Magnesium sulfate	N/A	Nitrile gloves, glasses
Fructose	N/A	Nitrile gloves, glasses
Sodium bicarbonate	N/A	Nitrile gloves, glasses
Sucrose	N/A	Nitrile gloves, glasses
Sodium carbonate		Nitrile gloves, glasses
Sodium hydroxide	 	Nitrile gloves, glasses
Potassium carbonate		Nitrile gloves, glasses
Boric acid	 	Nitrile gloves, glasses
Lithium chloride		Nitrile gloves, glasses

Iodine		Nitrile gloves, glasses
Phenolphthalein		Nitrile gloves, glasses
Copper sulfate		Nitrile gloves, glasses
Acetic acid	 	Nitrile gloves, glasses

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6.4: Procedure

General Instructions:

You will be testing and identifying four unknown white solids using the following qualitative scheme. Your lab instructor or lab assistant will give you your first sample. Record the **sample identification number** in the designated space on your data sheet. You will be able to test the substance directly by adding reagents that will cause it to react in a characteristic way. Proceed through the scheme until you have identified your unknown. When you have properly identified the first sample, return the vial to your lab assistant and you will be given the second sample. Repeat procedures for the second, third, and fourth samples. *Always wash and dry scoopulas or spatulas between samples.*

Place a "pea sized" amount of solid unknown (the amount that fits on the end of a scoopula or spatula) into a medium test tube each time a fresh sample of the unknown is indicated. Do not be wasteful. You may run out of your unknown.

Prepare a separate "Report Sheet" for each unknown sample.

Waste Disposal: All excess "unknowns" and products formed in this lab may be flushed down the drain.

Solubility Test:

1) Place a "pea-sized" sample (the amount of unknown that fits on the end of a scoopula, approx. 1/8 tsp.) in a clean medium-sized test tube. Add 5.00 mL DI H₂O.

2) Stir with a stirring rod or parafilm the tube and shake **gently**. Let stand about 1 minute.

3) If the substance **READILY** DISSOLVES, proceed to **PART B** of these procedures.

(Right side of flow chart.) If the substance does NOT readily dissolve, proceed to **PART A**. (Left side of flow chart.)

Part A:

Three water-insoluble white solids are possible: Cornstarch, CaSO₄, and CaCO₃.

1) TO TEST FOR CORNSTARCH: add 2 drops of 0.0500 M I₂ solution to your sample tube. Parafilm tube and shake gently.

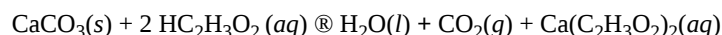
a) If a blue-black solution results, then cornstarch is present.

b) If a yellowish to brown solution results, then either CaSO₄ or CaCO₃ is possible.

2) TO TEST FOR CaCO₃: **place a fresh sample of unknown** into a clean test tube and add 5.00 mL DI H₂O. Parafilm tube and shake gently. Add 1.00 mL 6.0 M HC₂H₃O₂ (about 20 drops with a pasteur pipette). Parafilm and shake **gently**.

a) If bubbles evolve, then CaCO₃ is present. With concentrated solutions, bubbles of CO₂ gas evolve immediately. With dilute solutions, the effervescence will be much less obvious. (**Do NOT confuse "air bubbles" with the evolution of CO₂ gas.**) Warming in a water bath, with stirring, will increase the amount of bubble formation. CO₂ gas is odorless and colorless.

The molecular equation for this reaction is:



b) If bubbles do not evolve, then your substance must be CaSO₄.

Part B:

One of the eleven water soluble white solids is possibly NaOH, Na₂CO₃, K₂CO₃, MgSO₄, NaHCO₃, NaCl, LiCl, KCl, H₃BO₃, or sucrose.

1) Add 2-3 drops phenolphthalein indicator to your sample tube.

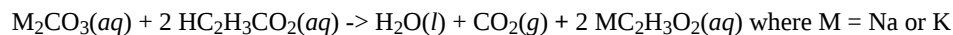
a) If the solution remains clear, proceed to **PART C** of these procedures.

b) If the solution turns pink, then NaOH, Na₂CO₃, or K₂CO₃ are possible, since they all form basic solutions when dissolved in water. Proceed to Step 2.

2) Place a **fresh sample** of your unknown in a clean test tube and add 5.00 mL DI H₂O. Stir with glass stirring rod until dissolved. Add 5-10 drops 6.0 M HC₂H₃O₂.

a) If bubbles do not evolve, then NaOH is the unknown present.

b) If bubbles evolve, then the evolution indicates the presence of either Na₂CO₃ or K₂CO₃. The molecular equation for this reaction is:



3) A **FLAME TEST** will distinguish the Na⁺ ion from the K⁺ ion.

a) Prepare a **fresh sample** of your unknown by placing a **double amount** in a clean test tube. Add a few mL DI H₂O and stir (does not have to dissolve). This will increase the solution concentration and give a more distinguishable color.

b) Light a bunsen burner and adjust the flame to produce a clear inner "cone".

c) Saturate the tip of a cotton swab with the unknown solution and place it into the hottest part of the flame. Observe the color of the flame:

Lavender-pink indicates the presence of the K⁺ ion and identifies K₂CO₃ as the unknown.

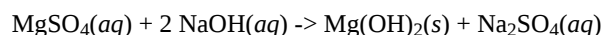
Bright yellow/orange indicates the presence of the Na⁺ ion and identifies Na₂CO₃ as the unknown.

Part C:

A clear solution remaining after the addition of phenolphthalein may contain one of the following compounds: NaCl, LiCl, KCl, NaHCO₃, H₃BO₃, MgSO₄, or sucrose

1.) **TO TEST FOR MgSO₄:** place a **fresh sample** of your unknown into a clean test tube and add 5.00 mL DI H₂O. Add 20 drops of 6.0 M NaOH. Parafilm and shake gently.

a) If a white precipitate forms, MgSO₄ is present. The molecular equation for this reaction is:



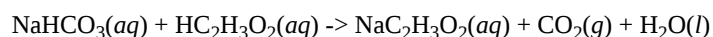
b) If no precipitate forms, then NaCl, LiCl, KCl, NaHCO₃, H₃BO₃, and sucrose are possible.

Part D:

LiCl, KCl, NaCl, NaHCO₃, H₃BO₃, and sucrose are possible.

1.) **TO TEST FOR NaHCO₃:** Place a **fresh** sample of unknown into a clean test tube. Add 5.00mL DI H₂O. Parafilm and shake gently. Add 10 drops of 6.0 M HC₂H₃O₂.

a) Evolution of bubbles indicates the presence of NaHCO₃. The molecular equation for this reaction is:



b) If no bubbles form, proceed to **PART E**.

Part E:

The remaining possibilities are sucrose, LiCl, KCl, NaCl, and H₃BO₃.

1) Prepare a **fresh sample** of the unknown by placing a *double amount* in a clean test tube. Add a few mL DI H₂O and stir (does not have to dissolve).

2) Perform a **FLAME TEST**. (see **PART B - 3**)

a) K⁺: lavender/pink unknown is KCl

b) Na⁺: bright yellow/orange unknown is NaCl

c) Li⁺: magenta unknown is LiCl

d) B³⁺: bright green unknown is H₃BO₃

e) no distinguishable color, but if the tip of the swab smells like “toasted marshmallows” (use a “wafting motion” to smell odor), then the unknown is sucrose.

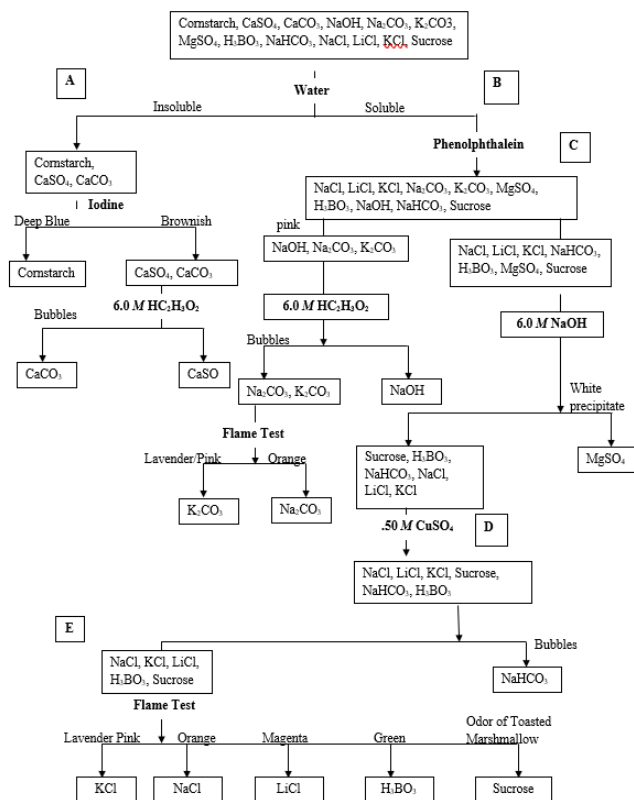
Data Sheet:

Lab ID: _____ Date: _____

Unknown Sample #: _____ Formula of Unknown Compound: _____

Treatment	Observation	Conclusion

Flow Chart:



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CHAPTER OVERVIEW

Experiment 7 - Determination of Acetic Acid

[7.1: Background](#)

[7.2: Pre-lab](#)

[7.3: Safety](#)

[7.4: Procedure](#)

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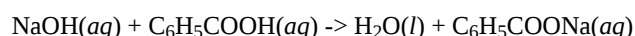
7.1: Background

Titration is the process of slowly and carefully adding one substance, known as the titrant, to another substance, known as the analyte, until stoichiometrically equivalent amounts have reacted. The amount of analyte present can be determined from the amount of titrant added. The substance referred to as the titrant is normally dispensed drop-wise from a piece of glassware known as a buret. The titrant is added to the sample until an indicator changes color. This color change occurs when enough titrant has been added to react with all the analyte. Titrations are used for chemical analyses because they are normally accurate, rapid, and inexpensive. The amount of acid or base present in a sample is often determined by titration.

In this lab, we will determine the amount of acetic acid in a vinegar sample. Vinegar is composed of mostly water and a small percentage of acetic acid. Sodium hydroxide is a common base that is used to acid titrations.

As an example, consider the titration of a sample of benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, (FW: 122.12 g/mol) with 0.1150 M NaOH. How much benzoic acid is in the sample if 25.35 mL of the base is required to neutralize the acid?

First, write a balanced equation for the reaction:



Second, determine the moles of titrant used:

$$0.02535 \text{ L NaOH} \times (0.1150 \text{ mol NaOH} / 1 \text{ L}) = 0.002915 \text{ mol NaOH}$$

Third, from the balanced equation determine the moles of acid equivalent to the moles of base used:

$$0.002915 \text{ mol NaOH} \times (1 \text{ mol C}_6\text{H}_5\text{COOH} / 1 \text{ mol NaOH}) = 0.002915 \text{ mol C}_6\text{H}_5\text{COOH}$$

Fourth, determine the weight of the acid present in the original sample:

$$0.002915 \text{ mol C}_6\text{H}_5\text{COOH} \times (122.12 \text{ g C}_6\text{H}_5\text{COOH} / 1 \text{ mol C}_6\text{H}_5\text{COOH}) = 0.3560 \text{ g C}_6\text{H}_5\text{COOH}$$

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7.2: Pre-lab

Week 1:

1. Calculate the molar mass of potassium hydrogen phthalate (KHP).

2. Write a balanced equation for the reaction between KHP and NaOH.

3. What is the concentration of a sodium hydroxide solution if 15.25 mL of the base was required to neutralize 0.4215 g of KHP?

Week 2:

1. Write the balanced equation for the neutralization of acetic acid with sodium hydroxide.

2. How many grams of acetic acid are in a sample solution if 22.50 mL of a 0.125 M sodium hydroxide solution are required to neutralize it?

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7.3: Safety

Chemicals	Hazards	Pictograms	Required PPE
Potassium hydrogen phthalate	N/A	N/A	Nitrile gloves, glasses
Sodium hydroxide	Corrosive to metals Skin/eye damage	 	Nitrile gloves, glasses
Phenolphthalein	Flammable liquid Causes serious eye irritation Suspected of causing genetic defects, cancer, damaging fertility or the unborn child May cause respiratory irritation, drowsiness, or dizziness		Nitrile gloves, glasses
Acetic acid	Corrosive to skin	 	Nitrile gloves, glasses

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7.4: Procedure

Week 1

Standardization of NaOH Solution

Commercial NaOH is not sufficiently pure to be used directly to make up a standard base solution. However, a solution of NaOH can be prepared and its molarity determined by titrating a known weight of a primary standard. A primary standard is a compound of high purity that can be weighed accurately. An acid that meets these requirements is potassium hydrogen phthalate, $\text{KHC}_8\text{H}_4\text{O}_4$.

This formula is commonly abbreviated as KHP. Note that the P here does not stand for phosphorus, but the phthalate ion, $\text{C}_8\text{H}_4\text{O}_4^{2-}$. You will use samples of KHP to standardize your solution of NaOH.

1. Preparation of desicc cooler

Check the desicc cooler in your cupboard. It may contain some of the drying agent, CaSO_4 , commonly called Drierite. The Drierite readily absorbs water and will maintain a very dry atmosphere in the desicc cooler when it is closed. If the indicating crystals are pink, then the Drierite must be replaced. If the indicating crystals are blue, then you may use your desicc cooler as it is. If you need to replace the Drierite, empty the desicc cooler into the bottle marked "USED DRIERITE". Add enough fresh Drierite to half-fill a 100 mL beaker, and then pour the Drierite into your desicc cooler. Finally add the desicc cooler plate for supporting bottles and replace the cover.

2. Drying of primary standard

The KHP must be dried before weighing to ensure accurate results. This has been done for you. Remove a bottle of the KHP from the oven. Place the bottle in your desicc cooler. Replace the desicc cooler lid and allow the bottle of KHP to cool for 15 to 20 minutes. Store the dried KHP in the desicc cooler until ready to weigh out the 3 samples for the titration.

3. Preparation of NaOH

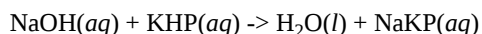
While waiting for the KHP to cool, prepare a NaOH solution that is approximately 0.1 M. Prepare the solution in a one liter plastic bottle. Since NaOH solutions etch glass, they are normally stored in plastic containers. Write the names of the students on a label and place the label on the bottle. To prepare the NaOH solution, use approximately 4.2 g of NaOH per 1000 mL DI water, or as specified by the laboratory instructor. Place the cap on the bottle and agitate the solution until all the NaOH pellets dissolve and the solution is thoroughly mixed. **Keep this solution until you finish section 5: Titration of Vinegar Samples.**

4. Standardization of NaOH solution with KHP

Make sure the analytical balance is zeroed. Using the analytical balance, weigh approximately 0.4 g KHP. Record the exact mass weighed. Transfer the KHP into a clean, numbered 250-mL Erlenmeyer flask. Repeat twice more, weighing out 3 samples of KHP for the 3 trials. **DO NOT WEIGH THE FLASKS.**

To each flask, add approximately 50 mL of deionized water, swirl to dissolve the KHP, and add 3 drops of phenolphthalein. Rinse the buret walls several times with a small amount of your NaOH solution. Fill the buret with the NaOH solution. Place a small beaker under the buret and fully open the stopcock to allow the buret tip to fill with solution. Be sure no air bubble remains in the top area when you zero the buret. Record the initial buret reading. Titrate the KHP sample in each flask in turn with NaOH solution to the first visible permanent pink color of the indicator. The final color should be light pink. Record the buret reading. Be sure to record both your initial buret reading and your final buret reading to two decimal places.

The reaction of NaOH and KHP may be represented by the equation



Data: Analysis of NaOH

Trial 1 Trial 2 Trial 3

1. Actual mass of KHP used _____ g

2. Formula weight KHP _____ g/mol

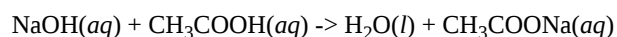
3. Moles KHP _____ mol
4. Moles NaOH _____ mol
5. Final buret reading _____ mL
6. Initial buret reading _____ mL
7. Volume NaOH used (mL) _____ mL
8. Volume NaOH used (L) _____ L
9. Molarity NaOH _____ M
- Average molarity NaOH _____ M

Week 2

Titration of vinegar samples

Obtain a 20-mL sample of vinegar. Rinse a 5-mL pipet with small portions of the sample, then transfer exact 5.00 mL portions to 3 Erlenmeyer flasks. Add about 75 mL deionized water and two drops of phenolphthalein indicator, and titrate with standard NaOH solution to the first faint but permanent pink color. Assume the density of the sample to be 1.004 g/mL, and calculate and report the percent acetic acid in the sample.

The reaction of NaOH and acetic acid may be represented by the following equation.



Data: Analysis of Vinegar

Trial 1 Trial 2 Trial 3

1. Final buret reading _____ mL
2. Initial buret reading _____ mL
3. Volume NaOH used (mL) _____ mL
4. Volume NaOH used (L) _____ L
5. Moles NaOH _____ mol
6. Moles CH₃COOH _____ mol
7. Molar mass CH₃COOH _____ g/mol
8. Mass CH₃COOH _____ g
9. Volume vinegar sample _____ mL
10. Mass vinegar sample _____ g
11. Weight % acetic acid _____ %

Average weight % acetic acid _____ %

Standard deviation weight %* _____

*standard deviation = range / sqrt(n)

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CHAPTER OVERVIEW

Experiment 8 - Chemical Kinetics

[8.1: Background](#)

[8.2: Pre-lab](#)

[8.3: Safety](#)

[8.4: Procedure](#)

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8.1: Background

Week 1

Chemical kinetics is the study of the rates and mechanisms of chemical reactions. The rate of a chemical reaction refers to the speed of the reaction. Some reactions, such as the rusting of iron, are very slow. Other reactions, such as the reaction of an acid and a base, are very fast. To determine the rate of a reaction, the time rate of change of the concentration of some reactant or product must be measured. One method of measuring reaction rates involves preparing solutions of known concentration and then measuring the time required for the concentration to decrease by a certain amount.

The ability to prepare a solution of accurately known concentration is an important skill in many disciplines. Chemists, biologists, pharmacists, and medical technologists are just a few examples of those who need to have this skill. Solution concentrations are often expressed in terms of molarity (M).

During Part 1 you will prepare solutions that will be used in Part 2 of the Chemical Kinetics experiment.

Week 2

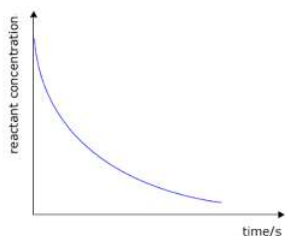
The rate of a chemical reaction refers to the speed of the reaction. Speed or rate is a concept familiar to everyone. For example, you can determine your average speed for a trip in an automobile by using the equation:

$$\text{Rate} = \text{change in odometer} / \text{time}$$

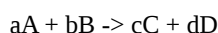
Using a similar method you can determine the rate of a chemical reaction except that now the rate is expressed as

$$\text{Rate} = \text{change in concentration} / \text{time}$$

The graph below shows the change in the concentration of a reactant with time. Note that the concentration decreases rapidly at first and then decreases more slowly. Reaction rates are not constant; they slow down as the reactants are consumed.



In general, the rate of a chemical reaction depends on the concentration of the reactants, but exactly how reactant concentrations affect the rate can only be determined by experiment. Consider the chemical reaction



where the capital letters represent chemical substances and the lower case letters represent the stoichiometric coefficients needed to balance the equation. The dependence of the reaction rate on the concentrations of A and B may be expressed by the "rate law" equation

$$\text{Rate} = k[A]^m[B]^n$$

where k is the rate constant and the square brackets mean "molar concentration". The exponents m and n must be determined by experiment. The rate constant is independent of reactant concentrations, but does depend on temperature. Changing the concentration of A or B will not affect k , but increasing the temperature or adding a catalyst will cause k and hence the reaction rate to increase.

The exponents can be determined by running a series of reactions with different concentrations of A and B and measuring the rates. Consider the experimental data in the table below:

<u>Experiment</u>	<u>A</u>	<u>B</u>	<u>Rate</u>
1	1.0 <i>M</i>	1.0 <i>M</i>	0.0050 <i>M/s</i>
2	2.0 <i>M</i>	1.0 <i>M</i>	0.0100 <i>M/s</i>
3	1.0 <i>M</i>	2.0 <i>M</i>	0.0200 <i>M/s</i>

By comparing experiments 1 and 2 we see that the concentration of B was the same and, therefore, had no effect on the rate. However, the concentration of A was doubled in the second experiment (2.0 *M* versus 1.0 *M*) and we note that the rate also doubled (0.0100 *M/s* versus 0.0050 *M/s*). This means that the rate is directly proportional to the concentration of A or mathematically the exponent $m = 1$. Similarly, if we compare experiments 1 and 3, where the concentrations of A are the same, we see that doubling the concentration of B (2.0 *M* versus 1.0 *M*) results in a four-fold increase in the rate (0.0200 *M/s* versus 0.0050 *M/s*). This means that exponent $n = 2$.

In general, if changing the concentration of a reactant has no effect on the rate, then the reaction is zeroth order (exponent = 0) in that reactant. If doubling the concentration doubles the rate, then the reaction is first order (exponent = 1). If doubling the concentration increases the rate by four, then the reaction is second order (exponent = 2).

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8.2: Pre-lab

Week 1:

Solve each of the following problems. Show all calculations in the space provided. Be careful about significant figures.

1. What mass of potassium iodide is required to prepare 100.0 mL of 0.200 M KI?

molar mass of KI = _____

mass of KI required = _____ (Also record result in Part B.)

2. What mass of potassium chloride is required to prepare 25.00 mL of 0.200 M KCl?

molar mass of KCl = _____

mass of KCl required = _____ (Also record result in Part B.)

3. To what final volume must 20.00 mL of a 0.200 M KI solution be diluted so that the final concentration is 0.0800 M KI?

final volume of KI solution = _____

Week 2:

Use the following data to determine the exponents for the reaction:

$aA + bB + cC \rightarrow \text{Products}$

The rate law for this reaction may be written as

$\text{Rate} = k [A]^m [B]^n [C]^p$

where k is the rate constant.

	[A]	[B]	[C]	Rate (M/s)
Trial 1	0.3	0.05	0.05	5.7×10^{-5}
Trial 2	0.3	0.1	0.05	5.7×10^{-5}
Trial 3	0.3	0.05	0.1	1.2×10^{-4}
Trial 4	0.4	0.05	0.2	3.1×10^{-4}
Trial 5	0.4	0.05	0.05	7.6×10^{-5}

A) What trials should you compare to determine the exponent for reactant A?

_____ and _____

By what factor does the rate change for these two trials? _____

What is the exponent for A? _____

B) What two trials should you compare to determine the exponent for reactant B?

_____ and _____

By what factor does the rate change for these two trials? _____

What is the exponent for B? _____

C) What two trials should you compare to determine the exponent for reactant C?

_____ and _____

By what factor does the rate change for these two trials? _____

What is the exponent for C? _____

Using the data for one of the trials, determine the value for the rate constant k . Show calculations below. Include units.

k = _____

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8.3: Safety

Compound	Hazards	Pictograms	Required PPE
Sodium chloride	N/A	N/A	Nitrile gloves, glasses
Potassium chloride	N/A	N/A	Nitrile gloves, glasses
Ammonium sulfate	N/A	N/A	Nitrile gloves, glasses
Starch	N/A	N/A	Nitrile gloves, glasses
Sodium thiosulfate	N/A	N/A	Nitrile gloves, glasses
Potassium iodide	Eye/skin irritation		Nitrile gloves, glasses
Ammonium persulfate	Skin/eye/respiratory irritation Oxidizer	  	Nitrile gloves, glasses
Copper sulfate	Skin/eye irritation Toxic to aquatic life	 	Nitrile gloves, glasses

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8.4: Procedure

Week 1:

You will be given the concentration and volume of the solutions you are to prepare. You will need to calculate the quantity of solute necessary to prepare the given volume of solution, then weigh out the appropriate amount of solute and dissolve to prepare the appropriate amount of solution. Prepare the solutions using the same technique as Lab 6. Show your calculations for determining the quantity of solute necessary to prepare the following solutions. (Remember to use correct units and significant figures.)

1. 100.0 mL of 0.200 M KI

mass of KI required to prepare 100.0 mL 0.200 M solution = _____

Prepare this solution and label. Use the top loading balance that weighs to the nearest 0.001 g.

2. 25.00 mL of 0.200 M KCl

mass of KCl required to prepare 25.00 mL of 0.200 M solution = _____

Prepare this solution and label. Use the top loading balance that weighs to the nearest 0.001 g.

The label for each solution should have the following information:

-concentration and chemical formula of substance

-name of substance

-date of preparation and your initials

Store these solutions in your lab desk drawer. You will use them in Part 2 of the experiment on chemical kinetics next week. Place the 25 mL volumetric flask inside a small beaker so that it cannot fall over.

Week 2:

1. In separate, labeled beakers, obtain approximately 100 mL each of KI and $(\text{NH}_4)_2\text{S}_2\text{O}_8$ and 60 mL of $\text{Na}_2\text{S}_2\text{O}_3$. Reactions 1 - 5 will be carried out as described in Table 1. Use separate 100 mL or 10 mL graduated cylinders for measuring each solution. Label all glassware.

2. Carefully measure the appropriate amount of KI and $(\text{NH}_4)_2\text{S}_2\text{O}_8$ for reaction 1 into two separate 50 mL Erlenmeyer flasks. Label the flasks with the appropriate solution names. Be careful not to cross contaminate before starting the reaction

TABLE 1

Solutions	0.200 M KI	0.100 M $(\text{NH}_4)_2\text{S}_2\text{O}_8$	0.200 M KCl	0.100 M $(\text{NH}_4)_2\text{SO}_4$	0.0050 M $\text{Na}_2\text{S}_2\text{O}_3$
Rxn 1	20.0 mL	20.0 mL	0	0	10.0 mL
Rxn 2	10.0 mL	20.0 mL	10.0 mL	0	10.0 mL
Rxn 3	20.0 mL	10.0 mL	0.0	10.0 mL	10.0 mL
Rxn 4	Repeat Rxn 1 with the addition of one drop of 0.1 M CuSO_4 to ammonium peroxydisulfate solution.				
Rxn 5	Repeat Rxn 1 but warm the reactants to approximately 10 degrees above room temperature prior to mixing.				

3. Dispense the $\text{Na}_2\text{S}_2\text{O}_3$, KCl, and $(\text{NH}_4)_2\text{SO}_4$ and 3-4 drops of starch solution into a 250 mL Erlenmeyer flask. This is your reaction flask. The KCl and $(\text{NH}_4)_2\text{SO}_4$ do not react but are used to maintain a constant ionic strength for all solutions.

4. Simultaneously pour the KI and $(\text{NH}_4)_2\text{S}_2\text{O}_8$ solutions into the 250 mL flask, start the timer, and mix thoroughly by swirling the flask. Place the flask on a white piece of paper to see the color change better. Note the time the reactants are mixed and measure the time elapsed until the solution turns blue-black or amber. Record the time in Table 2. Measure the final temperature of the solution and record in Table 2.

5. Repeat steps 3-5 for each of the different sets of reaction conditions specified in Table 1.

The concentration of iodide ion and peroxydisulfate ion in each reacting solution can be determined from the volume of each solution used and the final total volume (50.0 mL in each case). For example, 20.0 mL of 0.200 M KI when diluted to a volume of 50.0 mL gives an iodide concentration of 0.0800 M I^- . The concentrations of the reactants in each of the other reacting mixtures are determined in a similar manner and are tabulated in Table 2.

CALCULATIONS: Complete Table 2 as you complete calculations in A-L.

TABLE 2

Experiment	time (s)	Rate	I^- (M)	$S_2O_8^{2-}$ (M)	k	T ($^{\circ}C$)
1			0.0800	0.0400		
2			0.0400	0.0400		
3			0.0800	0.0200		
4			0.0800	0.0400		
5			0.0800	0.0400		

A. Calculate the reaction rate for each of your reaction mixtures and record in Table 2. (Include units).

B. What two experiments do you need to compare to determine how the iodide concentration affects the reaction rate?

Experiments _____ and _____

C. Why did you choose the two experiments in part B?

D. Using the appropriate data, determine the value of m in the rate law. Round your value of m to the nearest whole number.

$m =$ _____

E. What two experiments do you need to compare to determine how the peroxydisulfate concentration affects the reaction rate?

Experiments _____ and _____

F. Why did you choose the two experiments in part E?

G. Using your appropriate data, determine the value of n in the rate law. Round your value of n to the nearest whole number

$n =$ _____

H. Rewrite equation (3) using the values of m and n you have determined.

I. Rearrange the rate law in step H to solve for the rate constant k . Do not use numbers, only variables.

J. Determine the values of k for experiments 1, 2, and 3 and record in Table 2. (Include units)

K. How did the addition of the catalyst, Cu^{2+} , affect the rate of the reaction?

L. How did increasing the temperature affect the rate of the reaction?

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